

THEODOROS PSALLIDAS

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📍 Egaleo, Greece 🌐 Website in tpsallidas 🔗 theopsall 🎓 Scholar



EXPERIENCE

Senior Software Engineer

ProxyFoods

📅 February 2025 – Present 📍 Washington DC, USA

Senior Software Engineer at ProxyFoods AI, building an AI-powered food product formulation platform for CPG companies.

- Developed core formulation engine with nutritional analysis, cost calculations, sensory profiling, and mass-balance validation across a microservices architecture.
- Built and shipped an agentic recipe generation system using LangGraph with multi-agent orchestration, RAG over food-science knowledge, and human-in-the-loop ingredient matching.
- Designed cross-service authorization with OpenFGA (Google Zanzibar), multi-tenant RBAC, and shared library infrastructure (auth, tracing, logging, exceptions) consumed across all services.
- **Tech stack:** React 19, TypeScript, FastAPI, Python, LangGraph, OpenAI GPT, LangFuse, OpenFGA, SQLAlchemy, Azure SQL, Docker, Azure (B2C, AppService, Artifacts), OpenTelemetry

ML Engineer & Software Developer

MyTripMyWay

📅 June 2025 – March 2026 📍 Greece

ML Engineer and Software Developer at MyTripMyWay, an EU-funded (NextGenerationEU) persona-guided trip planning platform for Greek tourism.

- Designed and implemented a persona-based recommendation engine with six interpretable travel personas, direction-aware itinerary optimization, and spatial-temporal feasibility modeling via Google Maps & GraphHopper.
- Built full-stack features across a multi-service architecture — POI aggregation (grid-based Google Places/Foursquare scraping), booking & payment flows (Stripe), and an interactive map-based itinerary UI.
- **Tech stack:** Next.js, React, MobX, Tailwind CSS, Leaflet, FastAPI, SQLAlchemy, MariaDB, Docker, Google Maps API, Stripe

Full Stack Software Engineer

Behavioral Signals

📅 March 2024 – February 2025 📍 Greece

Full Stack Software Engineer at Behavioral Signals, working on intelligent call-center routing and audio deepfake detection.

- Delivered agent-matching features; improved analytics, monitoring, and scalability for real-time routing systems.
- Designed, built, and owned the audio deepfake detection application end-to-end.
- **Tech stack:** ReactJS, FastAPI, Django, Docker, Ansible, Azure, AWS, PostgreSQL, Milvus, MLflow

EDUCATION

Ph.D. Candidate, Computer Science

University of Thessaly

📅 April 2021 – Present

Research on machine-learning methods for automated video summarization.

M.Sc. in Data Science

University of Peloponnese

📅 September 2019 – May 2021

B.Sc. in Computer Science

University of Thessaly

📅 October 2013 – June 2018

CERTIFICATIONS

Back End Development and APIs

freeCodeCamp

📅 October 2022

Cisco Certified Network Associate (CCNA) Routing and Switching

University of Thessaly

📅 October 2016 – June 2017

ACHIEVEMENTS



Code IoT Hackathon

First place at “Code the IoT Hackathon of MaTHiSiS Project” competition at the University of West Attica

LANGUAGES

Greek (native) English (fluent)

Machine Learning Engineer

NCSR Demokritos

📅 December 2022 – February 2024 📍 Ayia Paraskevi, Greece

Machine Learning Engineer at Multimedia Analysis Group (MagCIL), Computational Intelligence Lab contributing to the Museek Project.

- Developed a music copyright monitoring system; designed inference infrastructure, containerized services with Docker, and deployed end-to-end pipelines to IoT/Raspberry Pi fleets using Balena.
- Migrated models from classical ML to deep learning; reduced PostgreSQL footprint by transitioning to FAISS-based vector search while improving detection accuracy and latency.
- **Tech stack:** ReactJS, FastAPI, PyTorch, FAISS, PostgreSQL, Docker, balena, AWS, PostgreSQL, MLflow

Lead Software Engineer

Optechain

📅 October 2020 – February 2024 📍 Argypoli, Greece

Lead Software Engineer at Optechain

- Led frontend (React, Angular) and backend (Node.js, C#) for digital signage (Tizen/webOS) and EV-charging stations.
- Ran Azure/Docker stack with MSSQL; managed device fleet updates (Balena/Mender) to improve rollout safety and uptime.

Machine Learning Engineer

NCSR Demokritos

📅 December 2019 – July 2022 📍 Ayia Paraskevi, Greece

Machine Learning Engineer at Computational Intelligence Lab (CIL) for the Enorasi Project.

- Built video summarization and multimodal analysis models (PyTorch, scikit-learn) and shipped them for the Enorasi project.
- Improved summarization quality through research/experimentation on model architectures and feature fusion.
- Extended the benchmark dataset with a more robust, less subjective labeling pipeline to reduce variance and improve evaluation.

PUBLICATIONS

📖 Books

- Pikramenos, George et al. (2020). *Dimensionality Reduction and Attention Mechanisms for Extracting Affective State from Sound Spectrograms*, pp. 27–45. DOI: 10.1007/978-3-030-66125-0_3.

📄 Journal Articles

- Mitsou, Alexandros, Antonia Petrogianni, et al. (2024). “A multi-modal dataset for electric guitar playing technique recognition”. In: *Data in Brief* 52, p. 109842. DOI: 10.1016/j.dib.2023.109842.
- Psallidas, Theodoros and Evaggelos Spyrou (2023). “Video summarization based on feature fusion and data augmentation”. In: *Computers* 12.9, p. 186. DOI: 10.3390/computers12090186.

- Psallidas, Theodoros, Panagiotis Koromilas, et al. (2021). "Multimodal Summarization of User-Generated Videos". In: *Applied Sciences* 11.11, p. 5260. DOI: 10.3390/app11115260.
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Conference Proceedings

- Mitsou, Alexandros, Dimitra-Christina C Koutsiou, et al. (2022). "ENORASI Assistive Computer Vision-based System for the Visually Impaired: A User Evaluation Study". In: *Proceedings of the 15th International Conference on Pervasive Technologies Related to Assistive Environments*, pp. 668–677. DOI: 10.1145/3529190.3534784.
- Psallidas, Theodoros, Evaggelos Spyrou, and Stavros J Perantonis (2022). "Summarization of User-Generated Videos Fusing Handcrafted and Deep Audiovisual Features". In: *2022 17th International Workshop on Semantic and Social Media Adaptation & Personalization (SMAP)*. IEEE, pp. 1–6. DOI: 10.1109/SMAP56125.2022.9941864.
- Psallidas, Theodoros, Michael D Vasilakakis, et al. (2022). "Multimodal Video Summarization based on Fuzzy Similarity Features". In: *2022 IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*. IEEE, pp. 1–5. DOI: 10.1109/IVMSP54334.2022.9816266.
- Psallidas, Theodoros, Alexander Mitsou, et al. (2020). "ARCHEO: A Dataset for Sound Event Detection in Areas of Touristic Interest". In: *2020 15th International Workshop on Semantic and Social Media Adaptation and Personalization (SMA)*. IEEE, pp. 1–6. DOI: 10.1109/SMAP49528.2020.9248467.